Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/19/2006
PI Name:	Foster, Russell Grant Ph.D.		
Project Title:	The Role and Characterization of Novel Photoreceptor Mechanisms Regulating Circadian Rhythms, Sleep, Body Temperature and Heart Rate		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	russell.foster@eye.ox.ac.uk	**Fax:**	FY +44(0) 1865 287664
PI Organization Type:	UNIVERSITY	**Phone:**	+44(0) 1865 287680
Organization Name:	University of Oxford		
PI Address 1:	Wellcome Trust Centre for Human Genetics		
PI Address 2:			
PI Web Page:			
City:	Oxford	**State:**	AK
Zip Code:	OX3 7BN	**Congressional District:**	0
Comments:	UNITED KINGDOM		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	01/01/2004	**End Date:**	12/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-HPF00301		
Performance Goal No.:			
Performance Goal Text:			
	The long-term aims of this project are to provide effective countermeasures for the detrimental effects of the irregular and altered patterns of light exposure in space and on the surface of other planets. These countermeasures will depend upon an understanding of the basic mechanisms of how the eye perceives light. The eye has been one of the best-studied parts of the central nervous system and its fundamental functions were considered well understood. All photoreception within the eye of mammals was considered the province of the rods and cones. However, studies over the past decade have shown that varied aspects of physiology and behavior are regulated by gross changes in environmental light, and that these irradiance changes are detected by novel ocular photoreceptors. The photoreceptive mechanisms underlying these non-rod, non-cone photoreceptors remain poorly characterized. Yet their importance in our daily lives is profound. It is now clear that light exposure can influence alertness and sleep propensity. Furthermore, light regulates the phase of circadian clocks, and thus the timing of rhythmic functions such as digestion, sleep and		

Task Description:	performance. The central aims of the research that we are undertaking are to characterize the molecular mechanism of this unexplored photoreceptor system of the eye, and determine the extent to which these photoreceptors contribute to varied aspects of physiology and behavior. In this regard our proposal spans 1-4 of the Countermeasures Readiness Levels (CRL) of NASA's Biomedical Research and Countermeasures Program. By employing a unique rodless and coneless mouse model (rd/rd cl), and taking advantage of the new opportunities created by a range of post-genomic technologies, including access to the mouse and human genomes, we will address two broad questions: Question 1: What is the role of novel photoreceptors in the regulation of general physiology and behavior? The experiments in this section will determine the extent to which body temperature, heart rate (ECG) and brain activity (EEG) are modulated by non-rod, non-cone ocular photoreceptors. Furthermore we will define the relationship between novel ocular photoreceptors, light, sleep state and levels of c-fos expression in the ventrolateral preoptic nucleus (VLPO) of the brain. Question 2: What are the molecular mechanisms of non-rod, non-cone ocular photoreceptors? Three broad strategies are being undertaken to look for these molecules based upon bioinformatics, proteomics, and microarray technology. Our aim is to identify a set of genes that: a) share sequence similarity with proteins known to be involved in photoreceptor/sensory cell function; b) are expressed in light sensitive cells of the inner retina; c) undergo post-translational modification and/or changes in expression following light exposure. We will subsequently confirm that these genes are expressed in candidate photoreceptive cells using in situ hybridization, single cell PCR from photosensitive retinal neurons, and the generation of suitable knockouts where appropriate. The results from these integrative studies will provide the mechanistic substrate for both targeted drug development aimed at the manipulation of human circadian rhythms, sleep, mood and performance, and the design of new lighting sources that are either highly effective in regulating these novel photoreceptors or leave them largely unstimulated. During the first 12 months of this project all the personnel, infrastructure and new equipment has been established. We are now in the process of collecting data. The results emerging have convinced us that the methodologies are working and that the experimental designs are sufficiently robust to allow statistically meaningful conclusions to be drawn. Our main results so far are:a) We provide the first physiological evidence that non-rod non-cone photoreceptors modulate sleep. b) Our preliminary data suggest that the acute regulation of heart rate by environmental irradiance arise from inputs from a time-dependent interaction between the classical rod and cone photoreceptor pathways and the non-rod, non-cone ocular photoreceptors. c) We have established microarray technology in the lboratory to study gene expression in the rodless+coneless eye. During the next 12 months we will build upon these observations and proceed to publication.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New approaches to address the clinical problems that either result from, or are affected by, inappropriate lighting conditions (e.g. depression, insomnia, reduced alertness and circadian maladjustment associated with shift work, blindness or jet-lag) will be critically dependent upon a detailed understanding of the basic mechanisms of the non-rod, non-cone ocular photoreceptors. Thus a major earth-based benefit of the work outlined in this proposal will be to provide basic inormation for the design of new lighting sources that can be used to modulate mood and performance, including depression. Furthermore we will provide the mechanistic substrate for targeted drug development aimed at the manipulation of human circadian rhythms, sleep, mood and performance. Another benefit will be a greater understanding of the G-coupled receptor superfamily. This large family of membrane receptors is a major target for drug development, but many details of how these receptors work remain to be elucidated. Our studies will clarify the distinction between photopigment opsins and the "opsin-like" receptors. A final benefit from this work will be in the field of retinal disease. It has been estimated that over half of blindness can be attributed to retinal genetic defects and the involvement of inner retinal dysfunction in promoting dystrophy is an area of growing interest. Consequently, the functions of novel genes within the inner retina that either cause or contribute to genetic pathologies of the eye are of considerable significance.
Task Progress:	During the first 12 months all the personnel, infrastructure and new equipment has been established. We are now in the process of collecting data. The results emerging have convinced us that the methodologies are working and that the experimental designs are sufficiently robust to allow statistically meaningful conclusions to be drawn. Academic: I have been asked to speak at the key international meetings in both circadian biology and vision/photoreception over the past 12 months. I also contribute regularly to national and international seminar programmes. As a member of the fellowship committee for National Endowment of Science, Technology and the Arts (NESTA) in the UK, I am able to help promote science and science education related to the NSBRI outreach. Media/popular science: I often contribute to radio and television programmes and have written/co-written articles for the Times; Times Higher Education Supplement; Daily Telegraph; Observer; Irish Times. My book "Rhythms of Life" (Profile Books in the UK; Yale University Press in the US) received very favourable reviews in leading publications including Nature, Science, Current Biology, News Week; TLS. Education: I often speak in schools, and have organised and contributed to meetings for the general public including the British Association for the Advancement of Science; DANA Brain Awareness Week at the Science Museum, London; Royal Institution). I have also developed science education projects with several organisations that work in schools.
Bibliography Type:	Description: (Last Updated: 03/02/2017)
Articles in Peer-reviewed Journals	Barnard AR, Appleford JM, Sekaran S, Chinthapalli K, Jenkins A, Seeliger M, Biel M, Humphries P, Douglas RH, Wenzel A, Foster RG, Hankins MW, Lucas RJ. " Residual photosensitivity in mice lacking both rod opsin and cone photoreceptor cyclic nucleotide gated channel 3 alpha subunit." Vis Neurosci. 2004 Sep-Oct;21(5):675-83. PMID: 15683556 , Sep-2004
Articles in Peer-reviewed Journals	Peirson SN, Bovee-Geurts PH, Lupi D, Jeffery G, DeGrip WJ, Foster RG. "Expression of the candidate circadian photopigment melanopsin (Opn4) in the mouse retinal pigment epithelium." Brain Res Mol Brain Res. 2004 Apr 7;123(1-2):132-5. PMID: 15046875 , Apr-2004
Books/Book Chapters	Foster RG, Kreitzman L. "Rhythms of life : the biological clocks that control the daily lives of every living thing ." London : Profile Books, 2004., Apr-2004